

Profile

Machine learning engineer working across vision-language models, agentic systems, model evaluation, and applied ML. I enjoy turning research ideas into tools that work in specialist domains, from oil & gas documents and manufacturing workflows to legal text and medical imaging. Strongest in Python/PyTorch, fine-tuning, evaluation design, and building with domain experts.

Skills

ML & LLMs Engineering	VLMs, LLM fine-tuning, LoRA, RLVR/GRPO, DPO, prompt optimization, agent evaluation, RAG. Python, PyTorch, Hugging Face, vLLM, SGLang, scikit-learn, XGBoost, LightGBM, Docker, Git, Linux, Rust.
Data & Systems Math	pandas, SciPy, Plotly, LangChain, Qdrant, Neo4j, LLM APIs, TPU/GPU training, LaTeX. Statistics, probability, stochastic processes, time series analysis, probabilistic ML, Bayesian analysis, R.

Work Experience

Viridien Machine Learning Engineer	2025/08 – Present
VLM document classification: Fine-tuned vision-language models for oil & gas technical documents using LoRA SFT, RLVR methods (GRPO/GSPO), and prompt optimization with ACE-Agent and DSPy. Improved F1 / accuracy from 0.47 / 0.74 to 0.78 / 0.85 at comparable model size. Agentic label QC: Built a QC system for noisy technical-document labels with domain experts. Converted expert review patterns into vision-enabled checks, domain “skills.md” knowledge, and a critic-validator loop to make label decisions easier to audit. Agent terminal-use training: Fine-tuned Qwen models on multi-GPU infrastructure with Harbor, SGLang, and AREAL to improve terminal-use behavior, in collaboration with CAMEL-AI.	
Autodesk Intern, Technology Consultant (AI/ML)	2024/06 – 2024/09
VLM benchmarking for CAM: Created a benchmark dataset for evaluating how vision-language models understand Computer-Aided Manufacturing workflows. The benchmark supported model selection for Fusion agents and gives the team a repeatable way to test new model releases.	
Business-Intelligence of Oriental Nations Corporation Ltd Intern, Algorithm Engineer	2023/06 – 2023/09
Manufacturing ML: Trained machine learning models for manufacturing applications using scikit-learn and XGBoost.	

Selected Competitions & Hackathons

Kaggle Hackathon Honorable Mention (8/321) - [Solution Write-up]	May 2026
Fine-tuned Gemma 2 2B Instruct for reasoning with a curated data mixture, LoRA reasoning SFT, and DPO based on the Delta Learning Hypothesis. Trained on Google TPU.	
Kaggle Competition Master (Ranked 842 worldwide; 1x Gold, 2x Silver).	Feb 2026
Kaggle Competition Gold Medal (13/3357) - [Solution Write-up]	Feb 2026
Built a Rust solver for irregular 2D bin packing, placing non-convex shapes into the smallest possible bounding square across hundreds of configurations. Combined simulated annealing, local search, and constraint relaxation over the 3-month competition.	
Kaggle Competition Silver Medal (19/1151) - Presented at NeurIPS 2025 Special Session [Solution Presentation]	Oct 2025
Built a program-synthesis agent with an evolutionary database to solve ARC-AGI tasks while optimizing for minimal code length.	
Mistral AI Award - Hack the Law Hackathon	Jun 2025
Built a Mixtral-based multi-agent system to extract, classify, and link regulated activities and entities from legal documents.	
1 st Place - Infosys Cambridge AI Centre Hackathon	May 2025
Built Memory Plus , an MCP memory layer for LLMs with a local vector store and RAG. (50+ GitHub stars; 8k downloads)	

Publications

SeLoRA: Self-Expanding LoRA for High-Quality and Efficient Medical Image Synthesis	Feb 2026
Expert Systems with Applications • Project Page	
Semi-Supervised Medical Image Segmentation via Knowledge Mining from Large Models	Jan 2026
arXiv Preprint • Accepted to IEEE International Symposium on Biomedical Imaging (ISBI)	

Education

University of Cambridge, MPhil in Data Intensive Science (Distinction)	2024/10 - 2025/07
Advanced study in statistics, machine learning, high-performance computing, and data-intensive scientific applications.	
University of Edinburgh, BA in Mathematics (First Class Honours)	2020/09 - 2024/05
Mathematics degree with emphasis on statistics, machine learning, and mathematical foundations for modelling.	